

AICR · OPEN PROBLEM

Simultaneous Inversion of a One-Way Function and a Random Permutation With a Collision Finder

The strong amplification conjecture behind Simon's oracle

Status. Statement: AI-written from Geoffroy Couteau, Pooya Farshim and Mohammad Mahmoody, *Black-Box Uselessness: Composing Separations in Cryptography*, IACR Cryptology ePrint Archive 2021, not yet checked by a human. Open: the paper's weaker version without the collision finder is settled only in the case where the hard function is itself a random oracle and is open for an arbitrary one-way function, so the version stated here, with the collision finder, is open in every case; what the paper does establish is the two consequences, that this conjecture implies both the distributional and the class-reduction relaxations of its main helpfulness conjecture.

Category. *Impossibility Results & Black-Box Separations.*

Conjecture (informal). *Fix any function family that is hard to invert, in the sense that no efficient attacker succeeds on a random input with probability better than some bound. Now independently sample a random permutation, and hand the attacker a further oracle that, given any circuit built out of the permutation and out of recursive calls to that same further oracle, returns a random collision for it. The attacker is handed two challenges at once, the image of a random input under the hard function and the image of a random input under the permutation, and must recover both inputs. The conjecture is that its success probability is at most the hardness bound for the function, multiplied by the probability of guessing a permutation preimage outright, up to a polynomial factor. In words: the two inversion tasks are so independent that the attacker can do no better than solve them separately, and the collision-finding oracle does not help. The paper’s weaker version of this statement, in which the collision finder is dropped, is known when the hard function is itself a random oracle but not when it is an arbitrary one-way function; the version with the collision finder is stated here as a conjecture and is not claimed to be known in any case. Beyond its own interest as a statement about random permutations, the conjecture matters because the collision-finding oracle is exactly Simon’s, the one used to separate collision-resistant hashing from one-way functions, and because a known construction turns any pair of simultaneously hard-to-invert functions into a collision-resistant hash function. Proving it therefore settles two weakenings of the question of whether one-way functions can help build collision resistance.*

1 The setting

Simon [Sim98] separated collision-resistant hash functions from one-way functions by exhibiting a random permutation RP together with a collision finder Coll^{RP} , relative to which one-way permutations exist but collision resistance does not. It is standard [IR89] that a random permutation cannot be inverted with probability better than $\text{poly}(\lambda)/2^\lambda$ for a measure-one choice of the permutation, and Simon’s analysis shows that this survives the presence of the collision finder, and indeed the presence of any independent auxiliary oracle.

The conjecture asks for the corresponding *simultaneous* state-

ment against an arbitrary independent hard function. The paper’s motivation is a route to its main helpfulness conjecture. Holmgren and Lombardi [HL18] show how to build collision-resistant hash functions in a black-box way from exponentially secure one-way product functions, i.e. tuples of functions no efficient adversary can invert simultaneously on random inputs with probability better than $\text{negl}(n)/2^n$; the paper notes in a footnote that their result as stated requires the two components to be equal and injective, with an extra-loss black-box reduction from an arbitrary pair. Taking one component to be an arbitrary one-way function F and the other to be RP , the conjecture supplies exactly the product hardness the construction needs, and so yields an auxiliary primitive, namely a random permutation packaged with a collision finder, from which no collision-resistant hash function can be built alone, by Simon’s separation, but from which one can be built once any one-way function is added. The paper proves that this route works for two relaxed notions of black-box reduction: one where reductions need only succeed for a measure-one set of implementations of the auxiliary primitive, and one where they are class reductions in the sense of Shaltiel [Sha20], restricted to the efficient one-way functions. The paper also flags a weaker version of the conjecture in which the collision finder is dropped, notes that it is known when F is itself a random oracle but not for an arbitrary one-way function, and says its study might be of interest beyond black-box helpfulness, for instance in hardness amplification. A footnote suggests an even weaker stepping stone: an ε -secure one-way permutation together with a plain random oracle. A second, independent route goes through the backdoored random oracle model of Bauer, Farshim and Mazaheri [BFM18], whose leakage oracle can implement Simon’s collision finder, conditional on a communication-complexity conjecture related to set intersection, and yielding in Section 6.4 the conclusion that a single backdoored random oracle, though black-box separated from collision-resistant hashing, is black-box helpful for it.

2 Notation and parameters

Throughout, $\lambda \in \mathbb{N}$ is the security parameter and PPT abbreviates probabilistic polynomial time. An *oracle-aided PPT machine* $A^{O_1, \dots, O_k}$

is a PPT machine with oracle access to $O_1, \dots, O_k$ that runs in probabilistic polynomial time for every choice of oracles, each oracle call counting as one step. We write $x \leftarrow S$ for a uniform draw from a finite set S .

Let Perm_λ denote the set of permutations of $\{0,1\}^\lambda$. We write $F = \{F_\lambda : \{0,1\}^\lambda \rightarrow \{0,1\}^*\}_{\lambda \in \mathbb{N}}$ for a family of functions regarded as a single oracle, $\text{RP} = \{\text{RP}_\lambda\}_{\lambda \in \mathbb{N}}$ with $\text{RP}_\lambda \in \text{Perm}_\lambda$ for a permutation family regarded as a single oracle, and Coll^{RP} for the collision finder attached to RP. The function $\varepsilon : \mathbb{N} \rightarrow [0,1]$ measures the hardness of inverting F , and q denotes a positive polynomial. We write π and Coll^π when referring to the same two objects as they are named in Simon's original separation.

The parameters of the statement are:

- λ , the security parameter, also the input length of both challenge functions.
- ε , the inversion hardness of the one-way function oracle; universally quantified over functions from the naturals to the unit interval.
- F , an arbitrary ε -secure one-way function family, independent of the permutation.
- RP , a uniformly random permutation family on bit strings of each length.
- Coll^{RP} , Simon's collision finder for circuits built from the permutation and from recursive calls to itself.
- A , the attacker, an oracle-aided PPT machine given all three oracles and both challenges.
- q , the polynomial slack in the conclusion, allowed to depend on the attacker.

3 The conjecture

Definition 1 (ε -secure one-way function oracle). Let $\varepsilon: \mathbb{N} \rightarrow [0, 1]$. A family $F = \{F_\lambda: \{0, 1\}^\lambda \rightarrow \{0, 1\}^*\}_{\lambda \in \mathbb{N}}$ is ε -secure one-way if for every oracle-aided PPT machine B and all sufficiently large $\lambda \in \mathbb{N}$,

$$\Pr_{x \leftarrow \{0, 1\}^\lambda} [F_\lambda(B^F(1^\lambda, F_\lambda(x))) = F_\lambda(x)] \leq \varepsilon(\lambda).$$

Definition 2 (Random permutation with Simon's collision finder). The *random permutation* $RP = \{RP_\lambda\}_{\lambda \in \mathbb{N}}$ is obtained by drawing $RP_\lambda \leftarrow \text{Perm}_\lambda$ independently for every λ ; the associated measure on families is the product of the uniform measures.

The *collision finder* Coll^{RP} is a randomised oracle defined by induction on the nesting depth of Coll-gates. On input a circuit C with RP -gates and Coll^{RP} -gates computing a function $\{0, 1\}^m \rightarrow \{0, 1\}^{m'}$, it returns a pair (u, u') obtained by drawing $u \leftarrow \{0, 1\}^m$ and then $u' \leftarrow \{v \in \{0, 1\}^m : C(v) = C(u)\}$, so that $C(u) = C(u')$. Fixing, for every circuit, the coins used in this sampling yields a deterministic collision finder for RP , and *with probability 1 over the choice of* Coll^{RP} refers to the induced measure on these deterministic choices. This is the oracle written Coll^π for a random permutation π in Simon's separation.

Conjecture 1 (Simultaneous inversion of F and RP given Coll^{RP}). Let $\varepsilon: \mathbb{N} \rightarrow [0, 1]$. For every family $F = \{F_\lambda\}_{\lambda \in \mathbb{N}}$ that is ε -secure one-way (Definition 1) and every oracle-aided PPT machine A , there exists a polynomial q such that, with probability 1 over the choice of the random permutation RP and of the collision finder

Coll^{RP} (Definition 2), for all sufficiently large $\lambda \in \mathbb{N}$,

$$\Pr \left[(x_1, x_2) = A^{\text{F,RP}, \text{Coll}^{\text{RP}}} (1^\lambda, F_\lambda(x_1), \text{RP}_\lambda(x_2)) \right] \leq \varepsilon(\lambda) \cdot \frac{q(\lambda)}{2^\lambda},$$

where the probability is taken over $(x_1, x_2) \leftarrow (\{0,1\}^\lambda)^2$ and the coins of A .

Remark 1 (Independence of the two challenges). Note that F is fixed before RP and Coll^{RP} are sampled, so it is independent of them.

Remark 2 (Openness). The paper does not prove the conjecture: “We did not manage to prove this conjecture, and leave it as an interesting open problem which might be of independent interest” (p. 8). Of the weak version without the collision finder it records that “The above conjecture is known in the case where F is itself a random oracle, but not when it is an arbitrary one-way function” (p. 31), and it introduces the version stated here with “In our context, though, we need an even stronger version, which we still deem plausible, where the random permutation comes with a collision finder Coll ” (p. 31). What the paper does prove are two consequences: Theorem 6.8, that the conjecture yields distributional black-box helpfulness of one-way functions for collision-resistant hash functions, and Theorem 6.12, that it yields class helpfulness over the class of efficient one-way functions, the latter via a Borel–Cantelli argument extracting a single auxiliary implementation good for all countably many efficient one-way functions. Both proofs are given as sketches, and both instantiate the pair of simultaneously hard functions as the given one-way function together with the random permutation, invoking Simon’s analysis to see that the permutation stays exponentially one-way in the presence of the collision finder and of any independent oracle.

Remark 3 (Caveats for a reviewer). The printed conjecture writes the first challenge as $P(x_1)$ rather than $F(x_1)$ in both the weak and the strong version. It is written here as $F(x_1)$, which is what the sur-

rounding prose and the proof of Theorem 6.8 require, but a reviewer should confirm this reading. Second, the printed display carries no quantifier over λ and leaves $\text{poly}(\lambda)$ unquantified; this has been read as “there is a polynomial q , depending on the attacker, such that the bound holds for all sufficiently large λ ”, and a reading with “for all λ ” would be formally stronger. Third, the conjecture as printed demands exact recovery of both preimages, $(x_1, x_2) = A(\dots)$, whereas the statement derived from it in the proof of Theorem 6.8 is the weaker any-preimage form; the paper’s exact-recovery form has been kept. Fourth, the definition of the collision finder above spells out the recursion by nesting depth and the induced measure on deterministic finders, details the paper leaves to [Sim98]; the intended object is Simon’s, and any discrepancy should be resolved in Simon’s favour. Finally, the post-2021 literature has not been checked exhaustively for a proof or a counterexample.

4 Bibliography

- [BFM18] Balthazar Bauer, Pooya Farshim, and Sogol Mazaheri. *Combiners for backdoored random oracles*. CRYPTO 2018, Part II, LNCS 10992, pages 272–302, Springer, 2018.
- [HL18] Justin Holmgren and Alex Lombardi. *Cryptographic hashing from strong one-way functions (or: One-way product functions and their applications)*. 59th Annual Symposium on Foundations of Computer Science, pages 850–858, IEEE Computer Society Press, 2018.
- [IR89] Russell Impagliazzo and Steven Rudich. *Limits on the provable consequences of one-way permutations*. 21st Annual ACM Symposium on Theory of Computing, pages 44–61, ACM Press, 1989.
- [Sha20] Ronen Shaltiel. *Is it possible to improve Yao’s XOR lemma using reductions that exploit the efficiency of their oracle?* APPROX/RANDOM 2020, Schloss Dagstuhl–Leibniz-Zentrum für Informatik, 2020.
- [Sim98] Daniel R. Simon. *Finding collisions on a one-way street: Can secure hash functions be based on general assumptions?* EUROCRYPT’98, LNCS 1403, pages 334–345, Springer, 1998.